

Revised Generation Interconnection System Impact Study Report

For Queue Project Q13-0250

Anza 230 kV
June 28, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q13-0250. The Interconnection Customer proposes a 460.1 MW solar photovoltaic facility interconnecting at the Anza 230 kV substation in the Illinois area. The request was received on July 01, 2013. The study evaluated system intact and single contingency conditions using the regional planning case, including 195 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$205,457,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ANZA 230-100 kV XFMR	ANZA to ASHLAN AVE 230 KV	336.7	135	100.0
ANZA to SKY VALLEY 100 KV Line	ANZA to ASHLAN AVE SUBSTATION 230 KV	181.5	142	48.3
KENZIE CRK to SOCRATES 19 100 KV	ASHLAN AVE to ANZA 230 KV L	176.4	111	38.7
KENZIE CREEK to OREGON CITY 100 KV	ASHLAN AVE to ANZA 230 KV	169.8	125	38.7
TOCOI to OREGON CITY 100 KV	ANZA to ASHLAN AVE 230 KV	169.6	127	38.7
TOCOI to SHAWBORO 100 KV	ANZA to ASHLAN AVE 230 KV line	169.3	128	38.7
GARDIAN to ANZA SUBSTATION 100 KV Line	ANZA to AHLAN AVE 230 KV	160.1	123	51.7
CARLYSS to BOCA RATON 230 KV L	CARLYSS to TOP CROP 500 KV line	155.3	560	65.4
HIGHGATE to TROY ENERG LLC 230 KV	TREGO to TROY ENERGY LLC 500 KV	148.3	534	13.2
STA S VAN NUYS to GOULD SUBSTATION 230 KV	TROY ENERGY LLC to RICHLAND 500 KV	130.8	511	7.0
BURWELL 500/345 kV Transformer Bank	ALLEGHENY ENERGY UNITS 3 4 & 5 345 kV/1 kV Auto	125.5	692	6.8
P.T.S to MEDORA SUBSTATION 345 KV	RITTER PARK to SUGAR HILL 345 KV	125.3	1,258	8.7
PINEY CREEK PROJECT SUBSTATION to CALDWELL 345 KV	HILLTOP SUBSTATION to PINEY CRK PROJECT SUBSTATION 500 KV	124.4	1,345	9.9
MASON ST 500/230 KV TX	TREGO to BEAVER 500 KV Line	124.1	1,081	12.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ALLEGHENY ENERGY UNITS 3 4 & 5 to SGAR HILL 345 KV	HILLTOP to PINEY CRK PROJECT 500 KV line	123.9	1,163	11.2
HAMPTON RD SUBSTATION to MEDORA 345 KV line	SIMPSON STUART TAP to FISHKILL PLAINS 345 KV	123.8	1,243	8.0
HIGHGATE to TREGO 230 KV	TREGO SUBSTATION to TROY ENERGYLLC SUBSTATION 500 KV	123.0	395	11.6
STEWART MONTAIN DAM to RITTER PARK 345 KV	MEDORA to P.T.S 345 KV	122.2	1,173	8.1
CHAMBERLIN to IMPERIAL 230 KV line	HYLEBOS to RAVENNA 500 KV	120.0	563	11.9
FISHKILL PLAINS to DILLARD-ORANGE 345 KV	TIDD to MUSKINGUM RIVER 500 KV	119.8	1,399	6.4
SKY VLY to SHAWBORO 100 KV	ASHLAN AVE to ANZA 230 KV	118.3	119	26.0
GALION GENERATING STATION 500 kV/230 kV Transformer	BEAER to TREGO 500 KV line	118.1	916	11.7
TALENENERGY MARTINS CREEK LLC HARRISBURG 500 kV/345 kV Autotransformer	CORONET to WESTWOOD GENERATION LLC 500 KV line	116.8	1,183	6.7
CARLEY ROAD to JAMES DE YOUNG 230 KV	E METHUEN 500/230 kV Transformer Bank	116.4	463	7.5
BURWELL to P.T.S 345 KV	ALLEGHENY ENERGY UNITS 3 4 & 5 345/1 kV Autotransformer	114.2	1,075	8.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ANZA 230/100kV TX	Replace transformer with higher-rated unit	\$7,922,000	100% (\$7,922,000)
SKY VLY SUBSTATION to ANZA 100 KV Line	Rebuild line with higher-capacity conductor	\$8,001,000	100% (\$8,001,000)
KENZIE CRK to SOCRATES 19 100 KV	Rebuild line with higher-capacity conductor	\$1,773,000	96% (\$1,699,000)
KENZIE CREEK to OREGON CITY 100 KV	Rebuild line with higher-capacity conductor	\$10,724,000	96% (\$10,273,000)
TOCOI to OREGON CITY 100 KV	Rebuild line with higher-capacity conductor	\$5,579,000	96% (\$5,344,000)
TOCOI to SHAWBORO 100 KV L	Rebuild line with higher-capacity conductor	\$5,333,000	96% (\$5,108,000)
GARDIAN to ANZA SUBSTATION 100 KV Line	Rebuild line with higher-capacity conductor	\$2,330,000	100% (\$2,330,000)
CARLYSS SUBSTATION to BOCA RATON 230 KV L	Rebuild line with higher-capacity conductor	\$89,392,000	92% (\$82,393,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
TROY ENERGY LLC to HIGHGATE 230 KV	Rebuild line with higher-capacity conductor	\$40,253,000	10% (\$4,011,000)
STA S VAN NUYS to GOULD SUBSTATION 230 KV	Rebuild line with higher-capacity conductor	\$22,038,000	10% (\$2,193,000)
BURWELL 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$30,901,000	10% (\$3,186,000)
P.T.S to MEDORA SUBSTATION 345 KV Line	Rebuild line with higher-capacity conductor	\$22,067,000	10% (\$2,279,000)
CALDWELL to PINEY CRK PROJECT 345 KV Line	Rebuild line with higher-capacity conductor	\$72,067,000	10% (\$7,440,000)
MASON ST 500/230 kV TX	Replace transformer with higher-rated unit	\$48,311,000	10% (\$4,829,000)
ALLEGHENY ENERGY UNITS 3 4 & 5 to SGAR HILL 345 KV	Rebuild line with higher-capacity conductor	\$17,727,000	10% (\$1,822,000)
HAMPTON RD SUBSTATION to MEDORA 345 KV line	Rebuild line with higher-capacity conductor	\$22,115,000	10% (\$2,285,000)
HIGHGATE to TREGO 230 KV	Rebuild line with higher-capacity conductor	\$37,244,000	10% (\$3,706,000)
STEWART MOUNTAIN DAM to RITTER PARK SUBSTATION 345 KV L	Rebuild line with higher-capacity conductor	\$22,740,000	10% (\$2,347,000)
CHAMBERLIN to IMPERIAL 230 KV line	Rebuild line with higher-capacity conductor	\$11,782,000	10% (\$1,189,000)
FISHKILL PLAINS to DILLARD-ORANGE 345 KV	Rebuild line with higher-capacity conductor	\$132,213,000	11% (\$14,023,000)
SKY VLY to SHAWBORO 100 KV	Rebuild line with higher-capacity conductor	\$24,260,000	96% (\$23,360,000)
GALION GENERATING STATION 500-230 kV TX	Replace transformer with higher-rated unit	\$26,829,000	10% (\$2,680,000)
TALENENERGY MARTINS CREEK LLC HARRISBURG 500 kV/345 kV Autotransformer	Replace transformer with higher-rated unit	\$29,793,000	10% (\$3,112,000)
CARLEY ROAD to JAMES DE YOUNG 230 KV	Rebuild line with higher-capacity conductor	\$25,706,000	10% (\$2,537,000)
BURWELL to P.T.S 345 KV line	Rebuild line with higher-capacity conductor	\$13,456,000	10% (\$1,388,000)
	Total allocated to project		\$205,457,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.